

EXPERIMENT 25. Multiple Lines with Legends, Widths and Colors

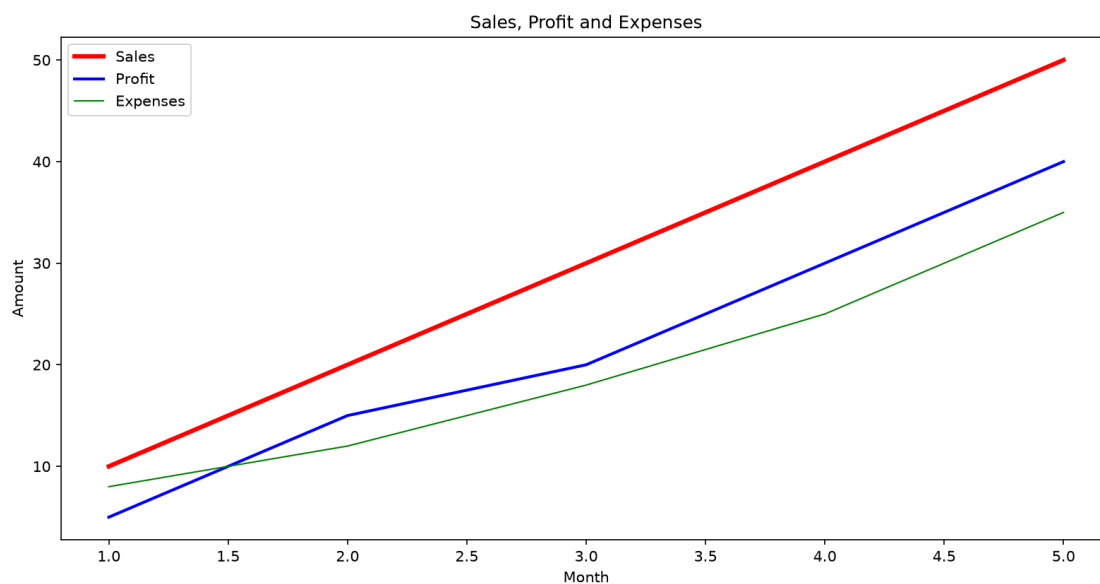
AIM:

To write a Python program to plot two or more lines with legends, different widths, and colors.

CODE:

```
exp25.py > ...
1  import matplotlib.pyplot as plt
2  x = [1, 2, 3, 4, 5]
3  sales = [10, 20, 30, 40, 50]
4  profit = [5, 15, 20, 30, 40]
5  expenses = [8, 12, 18, 25, 35]
6  plt.plot(x, sales, color='red', linewidth=3, label='Sales')
7  plt.plot(x, profit, color='blue', linewidth=2, label='Profit')
8  plt.plot(x, expenses, color='green', linewidth=1, label='Expenses')
9  plt.xlabel("Month")
10 plt.ylabel("Amount")
11 plt.title("Sales, Profit and Expenses")
12 plt.legend()
13 plt.show()
```

OUTPUT:



RESULT:

Thus, multiple lines were successfully plotted with different colors, widths, and legends.

EXPERIMENT 26. Create Multiple Plots

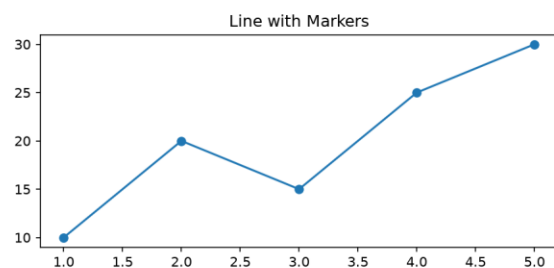
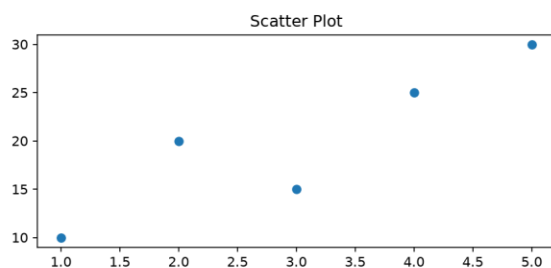
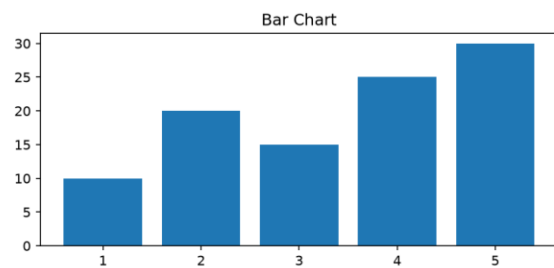
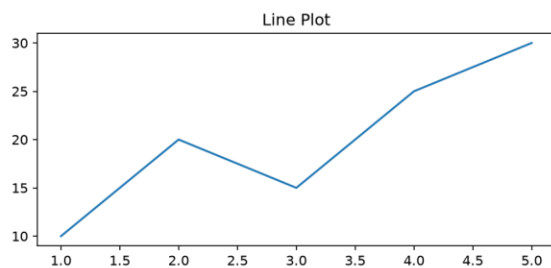
AIM:

To write a Python program to create multiple plots.

CODE:

```
exp26.py > ...
1  import matplotlib.pyplot as plt
2  x = [1, 2, 3, 4, 5]
3  y = [10, 20, 15, 25, 30]
4  plt.subplot(2, 2, 1)
5  plt.plot(x, y)
6  plt.title("Line Plot")
7  plt.subplot(2, 2, 2)
8  plt.bar(x, y)
9  plt.title("Bar Chart")
10 plt.subplot(2, 2, 3)
11 plt.scatter(x, y)
12 plt.title("Scatter Plot")
13 plt.subplot(2, 2, 4)
14 plt.plot(x, y, marker='o')
15 plt.title("Line with Markers")
16 plt.tight_layout()
17 plt.show()
```

OUTPUT:



RESULT:

Thus, multiple plots were successfully created using Matplotlib.